Progress Report and Results (round 2)

Sensor Trust for Lunar South Pole Landing - LANTERN testbed, 19 October 2026. Design and hypotheses were fixed in docs/PREREG.md (git tag prereg-round2) before the Site04 run.

Conclusions in plain words

- **The HAKUTO-R failure is real and reproducible.** A filter with a fixed rule and a coarse terrain reference rejects 99% of good altimeter readings on Site04 and the lander is lost in 499 of 500 runs.
- **The big fix is simply putting the terrain map into the filter.** Success goes from 0.2% to 82% (PITFAL-style) and to 97% with a plain map-tolerance gate that allows bigger surprises where the map is bumpy.
- **The four-step rule's own ingredients add little in the cliff cases.** Trust v2 (97%) matches the plain tolerance gate (97%); probation changes nothing (95.0% without it, 94.6% with it, v1).
- **Where trust v2 clearly wins:** a noisy altimeter in shadow (88% vs 64%), no map at all (91% vs 3%) and a mid-descent divert (91% vs 74%). The price: it is fooled by faults more often in the noisy case (52% vs 33%).
- **Where it fails badly:** a noisy altimeter in good light (39% vs 99.5% for PITFAL-style) and a map shifted by 2 km (1% vs 35%). Its physical-possibility check rejects readings that do not fit the map.
- **The camera lesson:** a navigation camera must never be used without its matcher's confidence (every fix used: 0.4-3% success). Predicting darkness from the Sun and terrain adds nothing over the matcher's own confidence (it is slightly worse).
- **Delayed readings:** the 83% for trust v2 comes entirely from time-stamping the readings. With the same time stamps the plain tolerance gate scores 97.6% and PITFAL-style 74%; without time stamps trust v2 scores 0%.
- **Round 1's headline (shadow-aware camera is the big win) did not survive a fair camera baseline.** That is the main scientific outcome of this round.

1. Verified in Unity

Item	Result
Descent	From 102.5 km to a flat Site04 spot: touchdown at t = 583-587 s, 0.50 m/s vertical, 0 m/s horizontal, 0.0 deg tilt, 251 kg fuel left, exactly on target
Live scan (LUMEN)	Fix from 2463 m down (patch cell grows with altitude). Error 47 m at 2205 m, 14.5 m at 921 m, 4 m at 155 m; median 8.9 m over 108 fixes. No fix below ~80 m
Replay scene	Site04Replay plays a recorded descent with the three filters side by side, with true lander, estimate ghost, altitude error, gate, camera trust bar and learned noise level

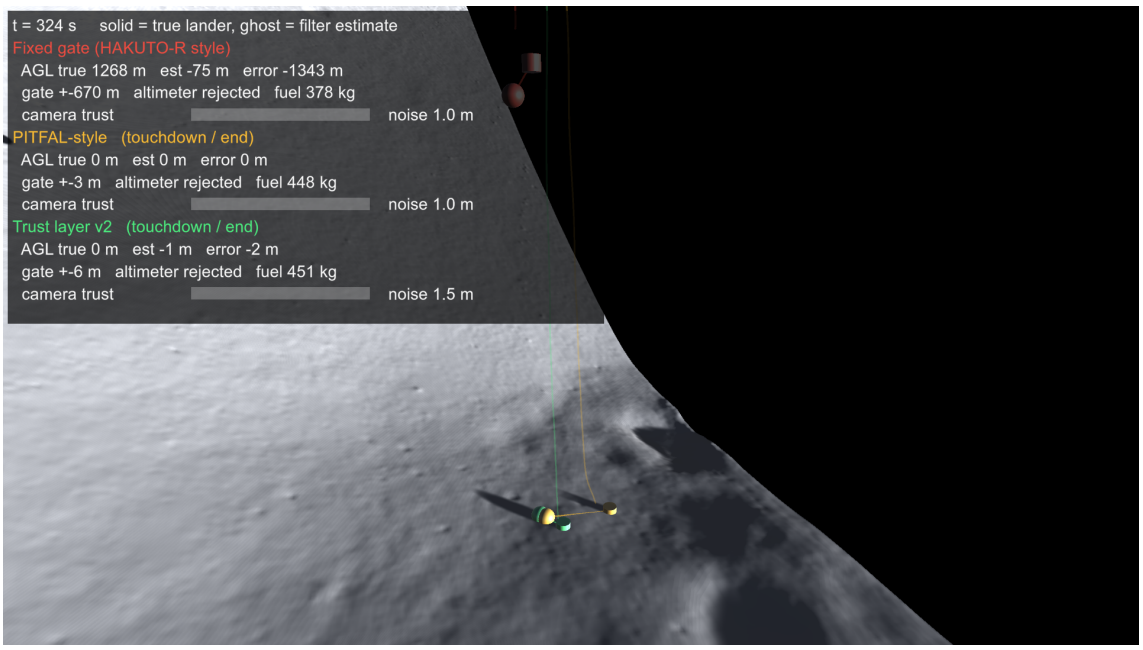


Figure 1. Unity replay of one descent (cliff + faults, run 7): the fixed-gate lander (red) hovers 1268 m up believing it is 75 m below the surface; the PITFAL-style (yellow) and trust v2 (green) landers have landed.

2. Camera reliability measured in Unity

300 paired nadir images on Site04 (true Sun vs a prediction from the map with 0.05-0.25 deg Sun-elevation error and +-1 deg azimuth error; reflections off so shadows are dark; read noise 2 DN plus shot noise), matched with a normalised-cross-correlation landmark matcher. The onboard prediction uses direct light only.

Lit fraction of footprint	Median error (m)	Fixes off by >100 m	Matcher confidence (median peak)
0 - 5%	401	88%	0.00
5 - 25%	7.6	20%	0.86
25 - 50%	2.4	0%	0.98
50 - 80%	1.3	0%	0.97
80 - 100%	0.8	0%	0.92

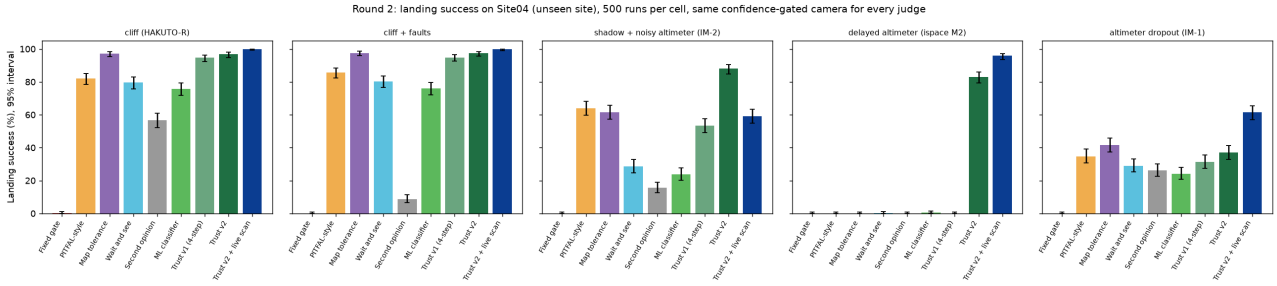
Table 1. Camera fix error by lit fraction. Among the 226 fixes with confidence >= 0.5, none was off by more than 100 m, so the matcher's own confidence is a reliable failure detector. This replaced round 1's hand-made damage curve. Caveat: the survey uses one renderer, one matcher and one site.

3. Site04 Monte Carlo (round 2, run once after the freeze)

500 seeded runs per cell, every judge uses the same confidence-gated camera. Probation window 10 samples and 4 sigma were tuned on procedural worlds and Site01 only (87% mean success there). The ML classifier was retrained on procedural worlds (held-out accuracy 89.5%, AUC 0.97). Sun 2 degrees above the horizon from the west.

Scenario	Fixed	PITFAL	Map tol.	Wait	2nd op.	ML	Trust v1	Trust v2	v2+scan
Cliff (HAKUTO-R)	0.2	82.0	97.2	79.6	56.6	75.8	94.6	96.8	100.0
Cliff + faults	0.0	85.6	97.6	80.2	8.6	76.0	94.8	97.4	99.8
Shadow + noisy altimeter (IM-2)	0.0	64.0	61.6	28.6	15.6	23.8	53.4	88.0	59.2
Delayed altimeter (ispace M2)	0.0	0.0	0.0	0.2	0.0	0.4	0.0	83.0*	95.8*
Altimeter dropout (IM-1)	0.0	34.8	41.6	29.0	26.2	24.2	31.4	37.0	61.4

Table 2. Landing success (%). 95% intervals about +-2 to 4 points. * Trust v2 uses time-stamped readings, the other judges do not (see the exploratory control).



4. Pre-registered hypotheses

#	Hypothesis (docs/PREREG.md)	Outcome
P1	Trust v2 beats PITFAL-style by >= 10 points in cliff, cliff + faults, shadow + noisy	Met: +14.8, +11.8, +24.0
P2	Lower false rejection than PITFAL-style in those scenarios	Mixed: lower in cliff (0.2 vs 0.8) and shadow + noisy (3.8 vs 4.3); higher in cliff + faults (1.7 vs 0.6)
P3	Fooled by faults at most 1 point more often	Met in cliff scenarios (4.9 vs 7.1, 6.1 vs 7.9); failed in shadow + noisy (52.4 vs 32.5)
P4	Sun-geometry camera prediction improves on the matcher confidence gate by <= 5 points	Supported: prediction is 5-11 points worse than the confidence gate in three of four cases and 2 points better in the fourth (trust v2, shadow + noisy)
P5	Gap to PITFAL-style on unseen Site04 at least the tuning-site gap minus 10 points	Met: Site04 98.0 vs 81.7 (cliff); Site01 100 vs 100
P6	Trust v2 lands >= 50% with delayed readings	Met numerically (83.0%), but only because of time stamps (exploratory control)

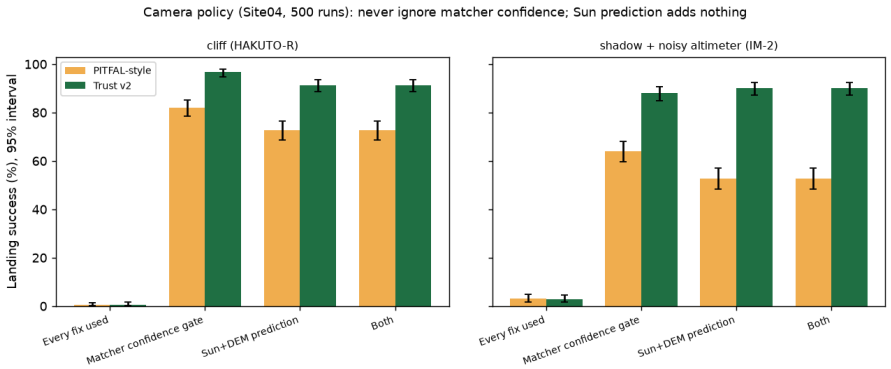
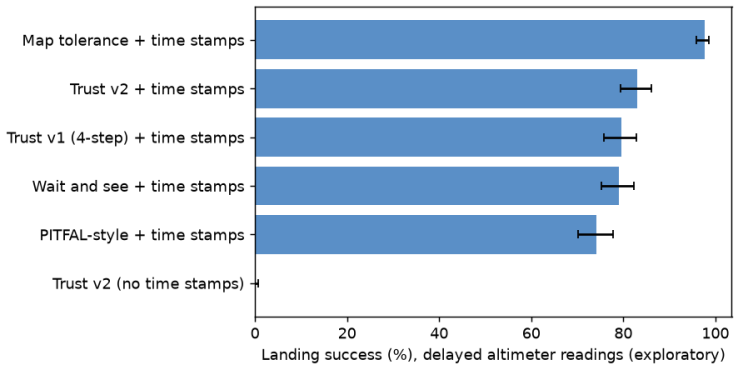


Figure 2. Camera policies (500 runs). Using every fix without a confidence check fails almost every landing; the matcher's own confidence is enough.

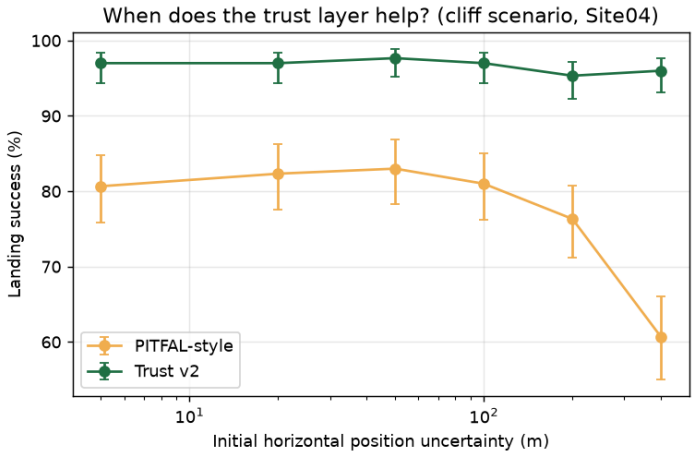
5. Exploratory control (not pre-registered)

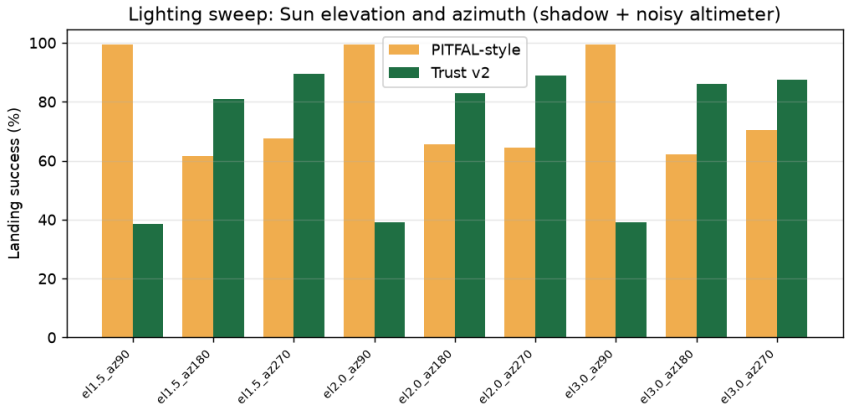
Does trust v2's delayed-reading result come from its trust logic or from time stamps? Every other judge was re-run with the same time stamps (500 runs): PITFAL-style 74.2%, wait-and-see 79.0%, trust v1 79.6%, trust v2 83.0%, **plain map-tolerance gate 97.6%**; trust v2 without time stamps 0.0%. The time stamps, not the trust rule, solve this scenario.



6. Sweeps and robustness (300 runs; sun sweep 200)

Test	PITFAL-style	Trust v2	Reading
Initial position error 5 / 50 / 200 / 400 m (cliff)	81 / 83 / 76 / 61	97 / 98 / 95 / 96	Gap grows with position uncertainty, as the plan expected
Mid-descent divert	74.3	91.3	Works without re-planning
No map	2.7	90.7	Fallback works; the baseline uses a coarse reference, so this shows the fallback design matters
Map shifted by 2 km	35.0	1.3	Trust v2 rejects readings that do not match the wrong map (90% false rejection)
Site01 (tuned on) vs Site04 (unseen), cliff	100 vs 81.7	100 vs 98.0	No drop on the unseen, harder site
Delayed readings, Site01 vs Site04	0 vs 0	98 vs 85	With time stamps
Sun from the east (well lit), shadow + noisy	99.5	39	Trust v2 collapses in good light with a noisy altimeter
Sun from the south / west, elevation 1.5-3 deg	62-70	81-90	Trust v2 ahead when the lander is in shadow





Live scan (extension): trust v2 + scan lifts cliff to 100%, delayed to 95.8% and dropout to 61.4%, but hurts shadow + noisy (59% with 203 of 500 runs timing out in a hover).

7. Limits and honest caveats

- Simulation only. The camera model comes from one renderer, one matcher and one site; shadowed scenes in the renderer are very dark but not physically exact.
- Disclosure: guidance, sensor settings and round-1 judge designs were chosen while looking at Site04 runs. Round 2 redid the camera model from measurements and fixed the settings in docs/PREREG.md, but a 100-run Site04 smoke test of the new code was seen before that document was written (recorded in it). The exploratory time-stamp control was added after the main run.

- Trust v2 assumes time-stamped altimeter readings and a map registered to within the position uncertainty; it fails when the map is wrong (1% with a 2 km shift).
- The no-map comparison is against a coarse-reference baseline; a terrain-relative baseline would be a fairer opponent.
- Site01 was used for tuning, so Site04 is the only unseen real site. The real HAKUTO-R altitude record was not available.
- The Python bench starts at 15 km as planned; the Unity demo starts at 102.5 km from rest and ignores orbital speed.

8. Is it publishable, and what next?

Yes as a controlled-comparison workshop paper (draft: docs/Paper_Draft.pdf): a measured camera model, a pre-registered design, and honest negative findings. The practical recommendation that falls out is a map-tolerance gate, a confidence-gated camera and time-stamped readings; the four-step rule is not needed for the cliff cases. Next: repair the two weaknesses found (physical check should scale with the noise level; wrong-map detection), add a terrain-relative no-map baseline, survey the camera on a second site and with a second matcher, and replay the real HAKUTO-R altitude record when it is obtained.

9. Repository map

Path	Contents
LANTERN/	Unity 6 project: Site04 and Site04Replay scenes, lander, autopilot, LUMEN, HUD, logger, replay, camera survey
sim/	Python test bench and pipeline (world, engine, pipeline, plots, camera survey scripts)
out/mc/	Round-2 results: main_results.csv, camera_policies.csv, sweeps_*.csv, exploratory_timestamps.csv, tuning.csv, ML report
out/mc_round1/, out/plots_round1/	Round-1 results (kept for comparison; git tag round1)
out/camera_survey/	Survey positions, matcher results, empirical camera model (images not in git)
docs/	PREREG.md, PROGRESS.md, this report, the project description, the paper draft